

Learner Guide

Hospital Admissions — what to do, why, and where you are in the process |

★★★★☆

This guide shows one sound way to complete the scenario, the reasoning behind each step, and which stage of the data analysis process each step belongs to. Assessors look for evidence that you followed the process in order; learners who jump straight to answers without working through the stages often lose marks. Work the task yourself first, then check your method here. Screenshots are illustrative mock-ups; Excel is shown, but Power Query or Python reach the same outcome.

The data analysis process — follow it in order:

Identify → Collect → Clean/Prepare → Analyse → Visualise → Interpret/Report

Each step below is tagged with its stage, so you can see the process being followed and evidence it to the assessor. Don't skip stages, and don't merge them — work through them in sequence.

Scenario 1 — Data Gathering and Validation

Step 1 — Identify what you need

STAGE 1 · IDENTIFY

What you do: Confirm the requirement before touching the tool: two data sources (`Admissions.csv` , `Hospitals.xlsx`) plus the Annex A BMI table, the common key that links them (**facid**), the known problem fields (**lengthofstay**, **pulse**), and the goal — one clean, merged, BMI-classified data set.

Why start here? Knowing the key, the problem fields and the end goal up front makes your cleaning and merge deliberate rather than trial-and-error, and shows the assessor you understood the task before manipulating data.

Step 2 — Bring each source in as a connection

STAGE 2 · COLLECT

What you do: Import each source as a **connection only** — don't load either to a sheet yet.

1. `Data` → `Get Data` → `From File` → `From Text/CSV` (`Admissions`) or `From Excel workbook` (`Hospitals`).
2. Select the file and click `Import` .
3. At the load prompt, choose `Only Create Connection` .

Import Data

×

Select how you want to view this data in your workbook.

☐ Table
 ☐ PivotTable Report
 ☐ PivotChart
 ☒ **Only Create Connection**

↑ Keeps the source as a query — nothing written to a sheet yet.

OK

Cancel

Figure 1 — Choose **Only Create Connection** so each source stays as a query.

Why connection only, not open the file? Opening a CSV gives a disconnected copy with no query and no refresh. Connection only keeps each source as a managed, refreshable query in the background, ready to clean and merge — and on a 10,000-row file it avoids cluttering the workbook with raw data you don't need on a sheet.

Step 3 — Double-click to open, then validate and clean

STAGE 3 · CLEAN / PREPARE

What you do: In **Queries & Connections**, double-click the Admissions query to open it in Power Query, then validate and correct the known problem fields:

- Sort or filter **pulse** to surface impossible values — there are two: a negative reading (-62) and an implausibly low one (27). Correct or remove them.
- Check **lengthofstay** for blanks — there are 27 empty records. Decide whether to remove them or exclude them from averages.
- Set each column's data type, and check for inconsistent formats (e.g. the `r count` value "5+", and the comma-formatted numbers in Hospitals).

Admissions — pulse column (sorted ascending to surface errors)

eid	gender	pulse	respiration	lengthofstay	
25	M	-62	6.5	4	⚠ impossible (negative)
1414	F	27	6.0	3	⚠ implausibly low
8123	F	58	6.5	2	
402	M	61	7.0	5	
77	F	63	6.5	1	

Sorting pulse low-to-high brings the two impossible values (-62 and 27) straight to the top.
Note also the 27 blank lengthofstay records to be addressed.

Figure 2 — Sorting pulse low-to-high brings the two impossible values straight to the top.

Why validate before anything else? A negative pulse or a blank length of stay will quietly distort every average and chart later. Catching them now — and showing you found them — is exactly what the "Validation" part of this scenario is testing. Sorting a column is the fastest way to make impossible values obvious.

Step 4 — Merge on the key, then Close & Load

STAGE 3 · CLEAN / PREPARE

What you do: Still in Power Query, **merge** the Admissions query with Hospitals on **facid** so each admission gains its hospital **Name**. Validate that all rows matched, then **Close & Load** to a table.

Why match on facid, not hospital name? The code is the reliable key — names can be spelt or formatted inconsistently, but facid (A–E) matches exactly every time. Merging only after cleaning means one correct, complete table is loaded for analysis.

Step 5 — Classify BMI

STAGE 3 · CLEAN / PREPARE

What you do: Add a **BMI Category** column using a nested IF against the Annex A bands:

```
=IF([@bmi]<18.5, "Underweight", IF([@bmi]<=24.9, "Healthy weight", IF([@bmi]<=29.9, "Overweight", IF([@bmi]<=39.9, "Obese", "Severely obese"))))
```

R2	fx	=IF([@bmi]<18.5, "Underweight", IF([@bmi]<=24.9, "Healthy weight", IF([@bmi]<=29.9, "Overweight", IF([@bmi]<=39.9, "Obese", "Severely obese"))))
eid	bmi	BMI Category
1	30.4	Obese
2	28.5	Overweight
3	23.1	Healthy weight

A nested IF assigns each patient to a band — a formula, so it updates if the data changes.

Figure 3 — A nested IF assigns each patient to a BMI band.

Why a formula, not manual labels? A formula applies the same rule to every one of the 10,000 rows consistently and updates if a BMI is corrected. Watch the band boundaries — 24.9, 29.9 and 39.9 are inclusive upper limits, so use `<=`, not `<`.

Scenario 2 — Data Analysis

Step 6 — Length of stay by hospital

STAGE 4 · ANALYSE

What you do: Group by hospital Name and calculate average, minimum and maximum **lengthofstay** (a PivotTable, or AVERAGEIFS / MINIFS / MAXIFS).

Expected result (cleaned set, 9,973 records):

Hospital	Avg LOS	Min	Max
North Valley General	3.24	1	13
Riverside Medical	3.32	1	12
West Town Health	4.77	1	12
Oceanview Hospital	5.00	1	13
Sunset Lakes Hospital	5.11	1	17

Step 7 — Visualise the comparison

STAGE 5 · VISUALISE

What you do: Present the hospital averages as a bar chart.

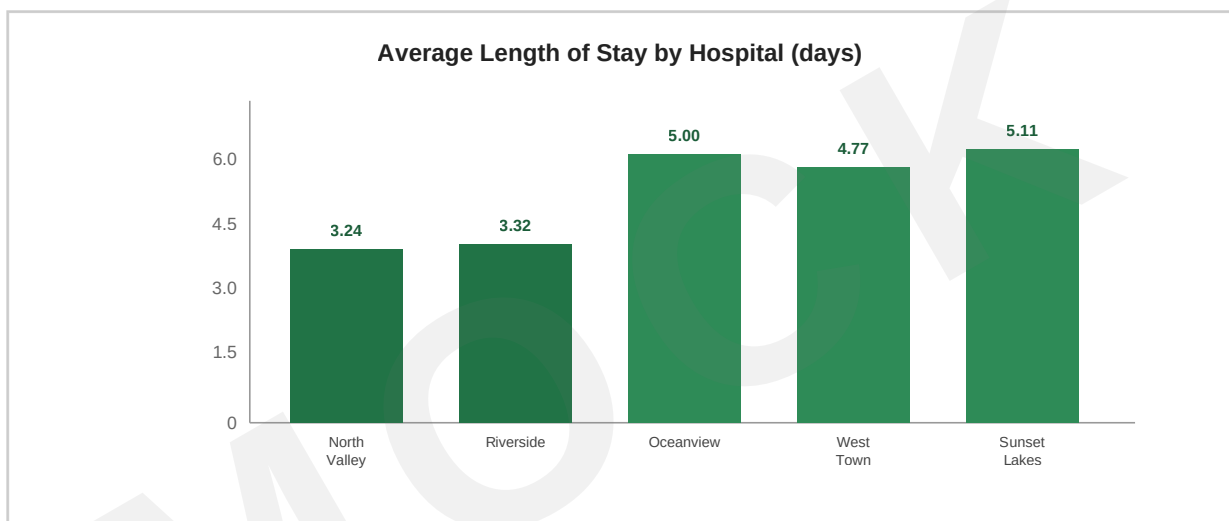


Figure 4 — Average length of stay by hospital.

Why visualise? A chart makes the split obvious instantly — two hospitals sit around 3 days and three sit around 5 — where a table of numbers takes longer to read. The visual carries the finding for a senior audience.

Step 8 — Length of stay by BMI and by gender

STAGE 4 · ANALYSE

What you do: Average length of stay for each BMI category, and for each gender.

Expected results:

By BMI category: Healthy weight **4.42**, Overweight **3.91**, Obese **4.04** days (almost all patients fall in the Overweight and Obese bands).

By gender: Female **3.87**, Male **4.12** days.

Step 9 — Explore other variables**STAGE 4 · ANALYSE**

What you do: Check whether lengthofstay relates to numerical fields such as **pulse** and **respiration**, using `=CORREL( . . . )` and a scatter plot with a trend line.

Why correlation and scatter? A correlation coefficient gives a single number for the strength of a relationship, and a scatter plot shows its shape and any outliers — together they tell you whether a variable is worth highlighting or is just noise.

Step 10 — Interpret and report**STAGE 6 · INTERPRET / REPORT**

What you do: Write a short summary for senior management beneath your charts, then save as `Admissions_Cleaned.xlsx`.

"After cleaning (two impossible pulse values corrected and 27 blank length-of-stay records removed), 9,973 admissions remain. Average length of stay splits clearly by hospital: North Valley and Riverside sit around 3.3 days, while Oceanview, West Town and Sunset Lakes sit around 5. Stay length varies little by BMI category (3.9–4.4 days) or gender (F 3.9, M 4.1), suggesting hospital is the stronger driver and warrants further investigation into why three sites run longer stays."

Marker's checklist — process followed in order

- **Identify:** sources, key (facid), problem fields and goal confirmed first.
- **Collect:** data brought in as **connection only** (not opened); double-clicked to open.
- **Clean/Prepare:** two bad pulse values corrected; 27 blank LOS handled; merged on facid; BMI classified by formula.
- **Analyse:** avg/min/max LOS by hospital, BMI and gender; relationships explored.
- **Visualise:** bar charts of the comparisons.
- **Interpret/Report:** findings summarised; saved as .xlsx.